

Ives (Hongqing) Xue

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Education

Wake Forest University, Winston-Salem, NC, May 2026

Bachelor of Science in Statistics, double major in Economics, minor in Mathematics

GPA: 3.8; Dean's List

Core Courses: Measure Theory (A), Real Analysis (A), Statistical Inference (A), Statistical Learning (A), Computational Statistics (A), Applied Markov Chain (In Progress)

Research Experience

Generalizable Classification under Label Shift

Winston Salem, NC

Honor Thesis Under Guidance of Dr. Ciaran Evans

Fall 2025 – Present

- Implemented BBSE & MLLS; benchmarked across label-shift regimes with random forest/logistic regression
- Compared distribution-distance metrics to guide model selection under shift
- Prototyping a generalizable pipeline: select shift-stable features and test robust models (incl. custom-loss neural nets)

Downwind Effects of Highways and Power Plants on EMS in NYC

Winston Salem, NC

Honor Thesis Under Guidance of Dr. Alex (Chu) Yu

Fall 2025 – Present

- Engineered day-level downwind exposure indices (wind $\times$ source geometry $\rightarrow$ ZIP-code EMS); highway index done, power-plant dataset in progress
- Specified three models (highway, plant, joint w/ interaction) to isolate effects, prioritizing measurement validity
- Analyzed DAC vulnerability and nonlinear/heterogeneous responses; justified wind/geometry exposures to mitigate bias from sparse monitors for credible EJ inference

Dirty Wind and EMS Demand: A Causal Forest Analysis

Winston Salem, NC

Summer Research Under Guidance of Dr. Alex (Chu) Yu

Spring 2025 – Present

- Built a monthly ZCTA panel (EMS—respiratory & total, ACS) with wind-based “dirty SW” exposure indices (continuous/binary/quantile)
- Used wind direction as quasi-experimental variation; fit 12 honest causal-forest models to estimate neighborhood $\times$ month CATEs and map heterogeneity
- Ran robustness checks (excl. Manhattan) and profiled more susceptible areas (rent burden, inequality, vehicle access, SNAP, single-parent share, race/ethnicity mix)

Mathematical Model for Syphilis with Reinfection

Winston Salem, NC

Individual Study Under Guidance of Dr. Chukwu

Spring 2024 – Fall 2024

- Built a compartmental ODE model for syphilis in MSM with S, I, latent, and two recovered states (treatment response vs. failure)
- Incorporated reinfection, treatment protocols, and temporary immunity with parameterized infection/treatment rates and waning
- Simulated dynamics with births, primary vs. reinfection, and treatment coverage to forecast disease progression

Collider Simulation

Winston Salem, NC

Undergraduate Research Under Guidance of Dr. McGowan

Spring 2024 – Summer 2024

- Extended replication: added baseline sodium, follow-up SBP, and early proteinuria; preserved temporal ordering
- Re-ran simulations using early proteinuria (vs. end-stage) to evaluate sodium→SBP dynamics
- Analyzed collider bias and documented risks/mitigation strategies

Other Experience

Research Assistant, Department of Economics, Wake Forest University, Spring 2024 – Present

- Automated R pipelines: cropland→panel; integrated temperature, crop maps, metadata
- Built Python ETL: user-day panel from 800+ files; mobility/app metrics
- Designed ZIP-level DAC classification; rebuilt FDNY EMS zones; normalized and matched 1M+ app titles

Tutor, Wake Forest Math and Stats Center, Wake Forest University, Fall 2023 – Present

Leadership Experience

President, CSSA, Wake Forest University, Spring 2025 – Present

Operation Chair, Chinese Ensemble, Wake Forest University, Fall 2022 – Present

Skills and Language

R, Python, Stata, Java, MATLAB, LaTeX, Maple, Microsoft Office

Bilingual in Chinese and English

Selected Competition and Fellowship

Wake Forest Research Fellowship (\$4000), Wake Forest University, 2025

Data Fest, Wake Forest University, 2024

Home Credit Challenge Bronze Prize, Kaggle, 2024